

Sequential Matching under Reciprocal Uncertainty: Ceilings, Acquisition, and Exact Matching

Round 1 Research Submission — Vouchsafe Sequential Matching Hackathon

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Abstract

We study the Sequential Matching Hackathon: twice daily for 60 days (+40 follow-up), a policy spends a 12-unit ask budget and proposes disjoint feasible pairs to maximise mutual second-meeting intentions (MSMI) per 100 arrived members, averaged over six scenario families. Our contribution is measurement-first. Only 1.24% of pairs are hard-feasible (0.24% sparse); one third of members can never be matched; 98.5% of every introduction is lost to mechanisms no policy controls, above all a 3-day answer window discarding about 64% of successes. The only reliable pair signal is agreement on 4 of 7 soft fields. The reachable edge set caps introductions, and greedy already realises 89–99% of the corrected ceiling — so this is an acquisition problem, not an RL problem (honest headroom about 1.3–1.6×). Our stdlib-only policy (matchability-gated VOI acquisition, exact general-graph maximum-weight matching, single-pass anchored online weights) climbs a paired ladder 0.333 → 0.514 (directional; the headline is underpowered about 20×). Fixed-prior beats adaptive everywhere, as noise theory predicts. All numbers come from 13 stdlib-only scripts against the public simulator; every person and outcome is synthetic.

1 Problem interpretation

Each of 60 decision days the policy (i) issues clarification asks costing at most 12 units (hard-constraint bundle = 3, one named soft field = 1; declined is permanent), then (ii) proposes a batch of pairwise-disjoint feasible pairs (empty = wait). Pairs must pass reciprocal hard checks both ways (age, gender, zones, smoking, children, structure, schedule), be unintroduced, with all hard fields known. Feedback matures with delay; 40 follow-up days resolve outcomes. MSMI needs both yes → date within 30 days → both second-yes within 3 days of the date. Score: mean over six families × 20 private seeds. Withheld: seeds, arrivals, hidden preferences, reply rates. Limits: JSON ≤ 1 MiB, 10 s/call, 2 cores, offline; memory self-carried.

Three facts reframe the challenge: (a) soft values are static (0 of 200 changed over 60 days) — asking is a lossless one-time purchase; drift shifts the scoring formula (−0.5 logit from day 35), not preferences. (b) Only 11.4% of pairs (3.8% cold start) show complete fit before asking, vs 56% after buying the four fields. (c) Only goal, pace, lifestyle and conversations enter the scorer (0.70/0.40/0.25/0.20; shifted: 0.25/0.80/−0.25/0.50); the other three soft fields are dead weight (force-changing them leaves MSMI bit-identical) and must never be bought.

2 Hypothesis

A reciprocal pair scorer learned online from delayed/censored feedback, with global maximum-weight allocation and VOI-ranked clarification, raises MSMI/100 over greedy via (i) learned weights vs uniform soft-count, (ii) global allocation vs greedy fill, (iii) soft acquisition vs none, (iv) drift-aware sequencing. Full RL is rejected: Learn2Match’s friction-loss result predicts failure exactly when volume is saturated and the gap is friction. LLM scoring is rejected (offline CPU; template conversations).

3 Reciprocal feasibility: introductions are capped

First-failing-rule attribution on truth: who-to-meet 57%, age 26%, geography 12% (19% sparse); the rest about 4% combined. $P(\text{matchable}) = 0.667/0.656/0.600$ — one third can never be matched. $|E|$ is a hard ceiling (no repeats). An apparent sparse contradiction ($|E| = 20.9$ vs 22.6 realised) was a seed-set mismatch: identical seeds give $|E| = 22.8$ vs 22.6 (99% saturated), assigned a subset of reachable with 0 violations. Clarification is supply-limited: 720 units vs about 80 askable members, exhausted by day 20. Sparse is saturated, not an opportunity; headroom is cold-start volume plus shift/drift ordering.

Family	Match	$ E $	Max	Greedy
dev*	133.4	103.1	34.5	~92 (89%)
sparse	131.2	20.9– 22.8	14.7	~99% †
cold start	120.0	90.5	31.8	77.6 (86%)

† Match = matchable/200; Max = max matching. Same-seed comparison. * +delayed/shift/drift.

4 Probability analysis (ranking only)

Per assigned pair: one shared shock $s \sim N(0, 0.45^2)$, then per direction $accept \sim \text{Bern}(\sigma(-0.25 + b_a + \text{fit} + s + \text{drift}))$, $respond \sim \text{Bern}(r_a)$; date w.p. 0.78; second yes $\sigma(0.15 + sb_a + s + 0.4[\text{goal match}])$. Directions correlate ($\rho \approx 0.43$ via fit + shared); never multiply marginals; integrate over s (41-node rule, error $< 5 \times 10^{-6}$). Replay validates ranking (top decile 0.041 pred / 0.033 emp vs bottom 0.002 / 0.000).

Funnel (greedy, development): 91.6 intros $\rightarrow$ 52.0 respond (57%) $\rightarrow$ 12.6 mutual (24%, the controllable stage) $\rightarrow$ 10.3 dates $\rightarrow$ 3.7 second-yes $\rightarrow$ 1.3 in-window $\rightarrow$ about 1.0 MSMI. The 3-day window $((3/5)^2)$ alone destroys about 64%. A draft “0.56 vs 1.0” under-prediction was a mislabelled single-batch sum; the corrected joint,

$$P_{\text{MSMI}} = r_1 r_2 (0.78)(0.36) E_s [A_a A_b S_a S_b]$$

with one expectation over the shared shock of all four sigmoids (acceptance $\times$ second-yes, both directions) and both response stages, gives a sum of about 1.07–1.15 vs about 1.0 realised — greedy at 87–93% of ceiling. Rankings are identical under either form. Top/bottom-half lift is $1.38 \times$ (fit) $\rightarrow$ $1.48 \times$ (+oracle rate), so member latents (1–2 obs/member) are not worth learning. Black-box weight search is ruled out by measurement: exploitable range 0.132 $<$ single-run noise 0.356. Common random numbers cut SE $1.7 \times$ and are mandatory.

5 Proposed policy (implemented, stdlib-only)

- (a) **Acquisition, matchability-gated.** v1 spent 183 of 384 soft asks (47.7%) on permanently unmatchable members — now gated out. Zone-clustered hard clearance first; leftover on the 4 live fields in tiers T0 (feasible pair) $\rightarrow$ T1 (cleared) $\rightarrow$ T2 (uncleared), by magnitude w_k , with partner-observed preference. Adaptive bar.
- (b) **Value model.** Observed both sides: $+w_k$ if equal else $-w_k/2$; unknown:

$$E[\text{contrib}_k] = w_k (1.5/N_k - 0.5), N_k \text{ options (exact here; kills the None-equals-None artefact).}$$

- (c) **Weights.** Single-pass streaming logistic on matured directional responses (v1 re-fed cumulative feedback, updating day-5 events about $55 \times$ — fixed via seen-event tracking); $\eta = 0.25/\sqrt{1+n}$, anchor $\lambda = 0.10$ to the development prior, clip 1.0. Synthetic check: pace 0.83 vs true 0.80.
- (d) **Allocation.** Exact general-graph MWM (branch-and-bound over components, greedy warm start; the graph is non-bipartite, about 47% same-gender acceptance); 268/268 fuzz-optimal vs brute force, < 1 ms/day. A -0.10 post-day-35 offset is a declared fixed parameter, not a drift detector (Round 2: trigger from the observed acceptance drop).

6 Missing and delayed data

Declined != not-asked (permanent, never bought). Missing responses are censored, never imputed; only matured outcomes update; pending queue keyed on expected observation day. Assignment-day feature snapshots (no leakage; static fields remove staleness). No IPS claims (propensity null; pool splits only; same-seed simulator comparisons). Memory about 10 KB. Thompson sampling over pool priors; rank-only decisions; posterior-predictive coverage check as the wrong-belief detector.

7 Evaluation

Baselines (identical seeds, all families): greedy, no-clarification, random-feasible. Ablations: no soft acquisition; fixed vs online weights; greedy-fill vs MWM; no drift scheduling; bandit off; threshold sweep. Measured ladder (seeds 101–106, paired):

Configuration	MSMI	vs	Mut
Organiser greedy	0.333	—	10.14
Candidate v1	0.347	+4.2%	10.56
VOI + greedy	0.389	+16.7%	10.50
Full (adaptive)	0.444	+33.3%	10.81
Full (fixed)	0.514	+54.2%	11.20

Honest reading: minimum detectable effect about 0.175 at $n = 36$, so only the fixed headline reaches it and per-variant cells ($n = 6$) are illustrative. The ladder is monotone, mutual acceptances rise 10%, and fixed beats adaptive even in shift (0.500 vs 0.417) — exactly as noise theory predicts (about 50 trials cannot carry 4 weights). Cost note: we spend about 409 vs about 198 ask units (tiebreaker loss; Round 2: cost-capped ablation). Harness smoke test (official evaluate.py, seed 101, development): greedy msmi = 1 $\rightarrow$ team msmi = 3, both valid, 0.15 s/call.

8 Failure cases and limitations

- Withheld answers (about 1/3 unmatchable, kept in denominators); saturated sparse supply; 8-day-busy schedule coupling (hence global matching); right-censoring (day-55 intros resolve about day 77, unseen by the policy).
- All outcomes synthetic — no evidence about real relationships. Kit vendored verbatim (pin a8e26b3, MIT + synthetic-data licence); our work is in research/; the team branch merged with history kept and policy files moved out of the kit subtree. Reproduce: unittest (22 OK) + verify_data.py, then the analysis scripts (prob_model, validate_model, why_cap, ceiling, value_spread, candidate, verify_team_policy, team_policy_eval).

References

- [1] Zong et al. Learn to Match. arXiv:2606.06744.
- [2] Li, Wang & Kong, bandit learning in matching markets survey, IJCAI 2025 (Liu-Mania-Jordan; Basu; Athanasopoulos et al. UAI 2026).
- [3] Udawani, non-adaptive greedy 1/2-competitive, arXiv:2403.18059.
- [4] Saar-Tsechansky et al., active feature-value acquisition, Mgmt Sci 2009; Ma et al., NeurIPS 2023.
- [5] Thompson (1933); Russo et al. (2018); Agarwal & Goyal (2013); Garivier & Moulines (2011); Joulani et al. (2013); Lakkaraju et al. (2017); Das & Kamenica (2005); Howard (1966).
- [6] Hayashi et al., off-policy evaluation for matching markets, RecSys 2025.
- [7] Segal-Halevi (2011); Drummond & Boutilier, IJCAI 2013; Blum et al., OR 68(2) 2020.

Appendix: reproducing this

```
cd The-Sequential-Matching-Problem
python3 -m unittest -v && python3 verify_data.py
cd ../research/analysis
python3 verify_team_policy.py
python3 team_policy_eval.py --seeds 101,102,103,104,105,106
```